

Imaging Microglia Surveillance during Sleep-wake Cycles in Freely Behaving Mice

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Xiaochun Gu , Zhong Zhao, Xueli Chen, Lifeng Zhang, Huaqiang Fang, Ting Zhao, Shenghong Ju, Weizheng Gao, Xiaoyu Qian, Xianhua Wang, Jue Zhang, Heping Cheng 

Jiangsu Key Laboratory of Molecular and Functional Imaging, Department of Radiology, Zhongda Hospital, Medical School, Southeast University, Nanjing 210009, China • Key Laboratory of Developmental Genes and Human Diseases, Department of Histology Embryology, Medical School, Southeast University, Nanjing 210009, China • Research Unit of Mitochondria in Brain Diseases, Chinese Academy of Medical Sciences, PKU-Nanjing Institute of Translational Medicine, Nanjing 211899, China • Institute of Basic Medical Sciences Chinese Academy of Medical Sciences, School of Basic Medicine Peking Union Medical College, Beijing 100005, China • Nanjing Raygenit Biotech Co., Ltd, Nanjing 211899, China • Academy of Advanced Interdisciplinary Study, College of Engineering, Peking University, Beijing 100871, China • National Biomedical Imaging Center, State Key Laboratory of Membrane Biology, Peking-Tsinghua Center for Life Sciences, College of Future Technology, Peking University, Beijing 100871, China

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Abstract

Microglia surveillance manifests itself as dynamic changes in cell morphology and functional remodeling in response to fluctuations in the neural environment. Whether and how microglia surveillance is coupled to brain state switches during natural sleep-wake cycles, as well as under sleep deprivation, remain unclear. To address this question, we used miniature two-photon microscopy (mTPM) to acquire time-lapse high-resolution microglia images of the somatosensory cortex, along with EEG/EMG recordings and behavioral video, in freely-behaving mice. We uncovered fast and robust brain state-dependent changes in microglia surveillance, occurring in parallel with sleep dynamics and early-onset phagocytic microglial contraction during sleep deprivation stress. With the aid of the biosensor GRAB_{NE2m}, we also detected local norepinephrine fluctuation occurring in a sleep state-dependent manner. We showed that the locus coeruleus-norepinephrine system, which is crucial to sleep homeostasis, is required for both sleep state-dependent and stress-induced microglial responses and β_2 -adrenergic receptor signaling plays a significant role in this process. These results provide direct evidence that microglial surveillance is exquisitely tuned to signals and stressors that regulate sleep dynamics and homeostasis so as to adjust its varied roles to complement those of neurons in the brain. *In vivo* imaging with mTPM in freely behaving animals, as demonstrated here, opens a new avenue for future investigation of microglia dynamics and sleep biology in freely behaving animals.

eLife assessment

This **important** study uses cutting-edge miniature two-photon microscopy to follow the structural dynamics of microglia in the somatosensory cortex of freely-moving mice across the sleep/wake cycle. **Solid** evidence revealed the brain-state-dependent regulation of microglial activity, highlighting alterations in microglial morphology during REM and NREM sleep phases compared to wakefulness. Furthermore, this study provides evidence for a critical role of norepinephrine from the locus coeruleus as a modulator of microglial morphology through the β 2-adrenergic receptor (b2AR). Overall, the article is an impressive technical feat to bridge a crucial gap in understanding sleep state-induced dynamics of microglia and its modulation by norepinephrine signaling.

Introduction

Sleep is a highly conserved and essential phenomenon of the brain and serves numerous cognitive as well as metabolic and immunologic functions. Moreover, the sleep-wake transition represents a specific and prominent change in the brain state (Besedovsky et al., 2019 [↗](#); Rasch and Born, 2013 [↗](#)). As compared to a wakeful brain, the sleeping brain shows more synchronous neuronal activities, which can be characterized by electroencephalography (EEG). Sleep has two main stages: the non-rapid eye movement (NREM) state, which is thought to promote clearance of metabolic waste partly through increased flow of interstitial fluid (Feng et al., 2019), and the rapid eye movement (REM) state, with EEG patterns similar to wakefulness yet with muscle atonia as well as vivid dreaming in humans (Besedovsky et al., 2019 [↗](#)). In addition, sleep, especially NREM, is directly involved in memory consolidation (Fogel and Smith, 2011 [↗](#); Rasch and Born, 2013 [↗](#)). However, the underlying biological dynamics and especially the cellular mechanisms of sleep regulation are not entirely understood (Frank, 2018 [↗](#); Frank and Heller, 2018 [↗](#)).

Growing evidence shows that microglia dynamics are coupled to and intertwined with the sleep-wake cycle (Deurveilher et al., 2021 [↗](#); Hristovska et al., 2022 [↗](#)). Microglia are resident innate immune cells ubiquitously distributed in the central nervous system and account for 10–15% of all brain cells (Wolf et al., 2017 [↗](#)). Microglia surveillance refers to dynamic changes in cell morphology that accompany functional remodeling in response to changes in the neural environment (Liu et al., 2019 [↗](#); Nimmerjahn et al., 2005 [↗](#); Stowell et al., 2019 [↗](#)). It has been shown that microglia ablation disrupts the maintenance of wakefulness and promotes NREM sleep (Corsi et al., 2022; Liu et al., 2021 [↗](#)), while enhancing an animal's fear memory consolidation (Wang et al., 2020 [↗](#)). Conversely, acute and chronic sleep deprivation (SD) leads to microglia activation amidst a proinflammatory state triggered by elevated circulating levels of cytokines (C-reactive protein, TNF- α , IL-1, and IL-6) (Krueger et al., 2011 [↗](#)). Thus, it is of critical interest to explore whether microglia surveillance can sense and respond to brain state changes during both natural sleep-wake cycles as well as under SD stress.

The investigation of microglial surveillance during the sleep-wake cycle demands technologies suitable to track the rapid dynamics of natural microglia behavior in real time. Though there have been pioneering studies on microglia dynamics in cultured brain slices (Honda et al., 2001 [↗](#)), static *ex vivo* conditions can hardly recapitulate the dynamic microenvironments and neuronal activities *in vivo*. Based on results obtained from fixed brain slices, it has been shown that chronic sleep restriction but not acute sleep loss causes microglia process changes (Bellesi et al., 2017), but it is difficult to use this approach to reconstruct the dynamic behavior of microglia accurately and quantitatively. As a major advance, two-photon microscopy has recently been applied to image

microglia in head-fixed or anesthetized animals, revealing microglial process motility, chemotaxis and homeostatic translocation, and multifaceted microglia-neuron interactions *in vivo* (Davalos et al., 2005 [↗](#); Eyo et al., 2018 [↗](#); Liu et al., 2019 [↗](#); Stowell et al., 2019 [↗](#)). However, because of mechanical constraints or anesthetics, this technology still precludes experimental paradigms in which natural sleep-wake cycle is undisturbed. In this regard, the recent advent of miniaturized two-photon microscopy (mTPM) has provided a powerful new tool ideal for fast and high-resolution brain imaging in freely behaving rodents (Zong et al., 2021 [↗](#), 2017 [↗](#)). With the aid of mTPM, researchers have mapped the functional network topography of the medial entorhinal cortex (Obenhaus et al., 2022 [↗](#)), deciphered the microcircuit dynamics in the dorsomedial prefrontal cortex during social competition (Zhang et al., 2022 [↗](#)), and unraveled the specific itch signal processing in the primary somatosensory cortex (Chen et al., 2021 [↗](#)).

In this study, we aimed to determine whether microglia surveillance undergoes any significant changes during natural sleep-wake cycles, and if so, what is the underlying regulatory mechanism. Using different models of mTPM tailored for high-resolution, multi-plane or large field-of-view (FOV) imaging, we monitored microglia in the somatosensory cortex of mice during natural sleep-wake cycles or while being subjected to acute SD. We found robust sleep state-dependent and SD-induced changes in microglia surveillance with characteristics differing from those induced by anesthetics. Furthermore, we showed that norepinephrine signals from the axonal projections of the locus coeruleus (LC) underlie the state-dependence of microglia surveillance and β_2 -adrenergic receptor (β_2 AR) signaling plays a significant role in this process.

Results

Imaging microglia dynamics in freely behaving mice

To visualize microglial morphology and dynamics during the sleep-wake cycle, microglia were specifically labeled with green fluorescent protein (GFP) expressed under the control of an endogenous Cx3cr1 promoter (Cx3cr1-GFP). Time-lapse imaging of the somatosensory cortex utilized the fast and high-resolution miniature two-photon microscope (FHIRM-TPM) recently developed by our laboratory (Zong et al., 2017 [↗](#)), and EEG/Electromyography (EMG) signals and behavior videos were simultaneously recorded to determine corresponding sleep-wake states (Fig. 1a-c [↗](#) and Supplementary Fig. 1). The focal plane of the FHIRM-TPM was placed 50-100 μ m beneath the surface of the somatosensory cortex with a field-of-view (FOV) of $220 \times 220 \mu$ m (Fig. 1d [↗](#)). Under the experimental conditions, we were able to track morphological and positional changes of a cohort of 5-10 microglia at a frame rate of 5 frames per second (FPS) over prolonged durations of greater than 10 hours (Supplementary Fig. 1 and Video), while the animals were allowed to roam and behave freely and their sleep-wake cycles remained undisturbed. Though we were initially concerned with potential phototoxicity associated with prolonged mTPM imaging, no changes in microglial morphology and fluorescent intensity were evident even after non-stop recording over the entire protocol period (Supplementary Fig. 1). This remarkable result indicates that photo damage is minimal, if any, during *in vivo* imaging (Supplementary Fig. 1).

Dynamic changes in microglial area, branching points, and process end-point speed were then analyzed using Imaris software (Fig. 1e1-e3' [↗](#) and Supplementary Fig. 1). We found that individual microglia maintained a relatively stable territory of surveillance over the entire experimental period, even while their individual projections extended and contracted incessantly, giving rise to a process end point motility of $1.39 \pm 0.09 \mu$ m/min in the wake state (Fig. 1f-h [↗](#)). These results show that mTPM in conjunction with multi-modal recordings of EEG/EMG and behavioral videos enables *in vivo* visualization of microglia dynamics over multiple time scales in freely behaving mice. Interestingly, we also observed occasional short-range translocation in a few cells, suggesting the ability of microglia to survey territories beyond their own boundaries, in general agreement with recent observations in head-fixed mice (Eyo et al., 2018 [↗](#)).

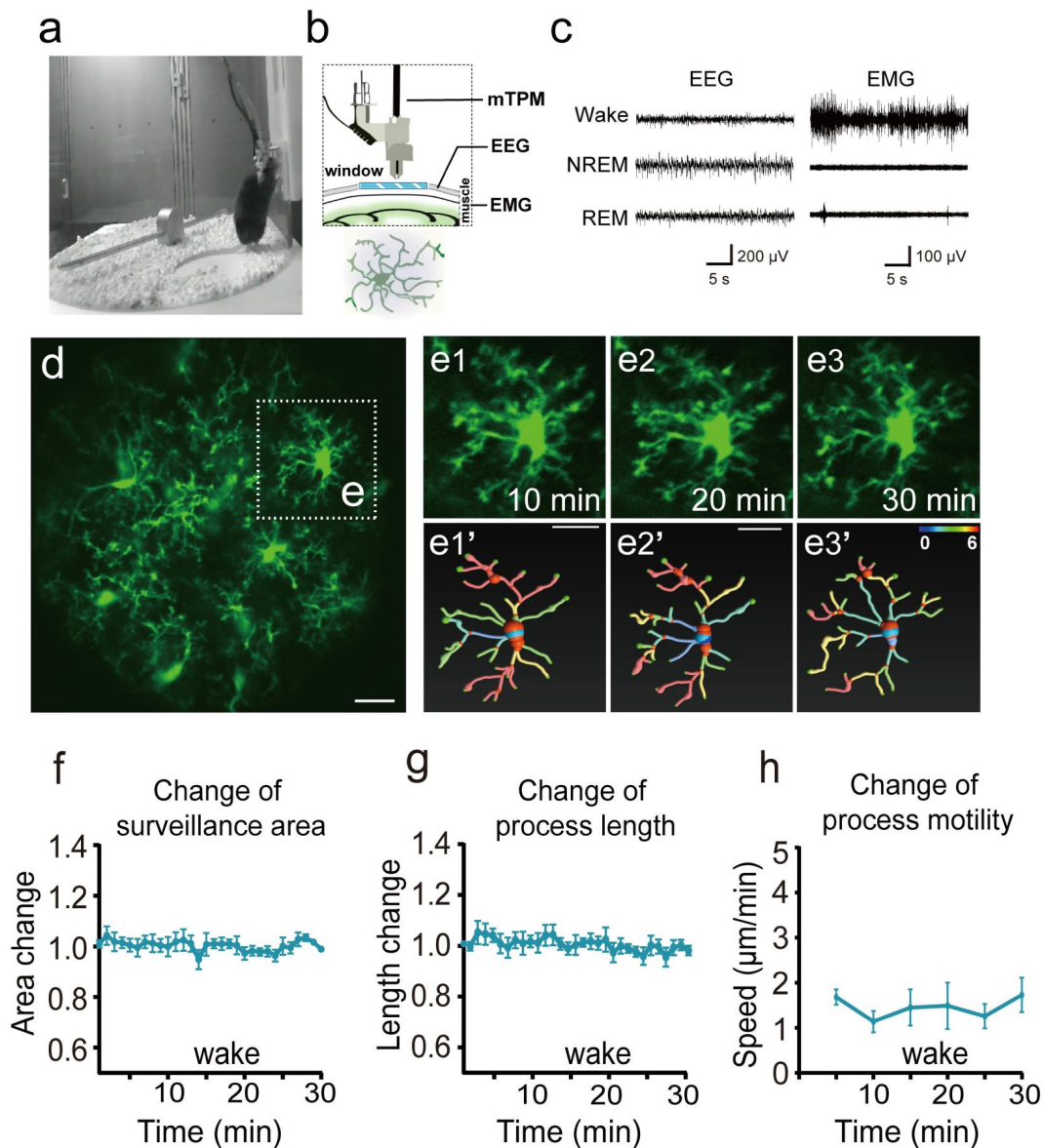


Fig. 1.

Imaging microglial surveillance in the somatosensory cortex in freely moving mice.

(a-c) Experimental setup. The animal was head-mounted with a miniature two-photon microscope (mTPM) and EEG/EMG electrodes and behaved freely in a cylindrical chamber (a). Microglia expressing GFP in the somatosensory cortex were imaged through a cranial window using the mTPM (b) and the sleep/wake state of the animal was simultaneously monitored using an EEG/EMG recording system (b-c).

(d-e) Microglial morphological dynamics when the animal was awake. A representative image with a FOV of $220 \times 220 \mu\text{m}^2$.

(e1-e3) Expanded views of selected microglia in box from (d) at 10 min (e1), 20 min (e2) and 30 min (e3) of continuous recording. **(e1'-e3')** Microglial process graphs digitally reconstruction for E1-E3 using Imaris software. In Figure e1'-e3', the branch points of processes are represented in red and the number of different branches of the whole cell is represented as gradient colors.

(f-h) Quantitative analysis of changes in microglia surveillance area (f), process length (g), and process motility, indexed by the speed of the extension and retraction at end points of the processes (h). Note that the gross morphology of microglia remained largely unchanged over a 30-min time frame in the wake state, despite significant motility at the ends of the processes. Scale bars, 25 μ m.

$n = 12$ cells from 3 mice. Scale bars, 30 μ m. See supplementary Fig. 1-3 [for more details](#).

Microglia surveillance during the sleep-wake cycle

By simultaneously imaging microglia and recording EEG/EMG throughout a sleep-wake cycle, we obtained continuous 4-6 h datasets from different mice and, in conjunction with video interpretation, segmented and sorted them into subgroups corresponding to wake, NREM, and REM states (**Fig. 2a** [↗](#)). For each state we quantified morphological features and process motility from 15-20 cells. Wake and NREM states lasting longer than 1 min were identified, for which the last 30-seconds recording was selected, and REM states lasting longer than 30 seconds for the characterization of possible state-dependent microglia dynamic surveillance.

We found that during the NREM and REM states, microglial process length and surveillance area increased compared with those during wakefulness (**Fig. 2b-d** [↗](#), b'-d'). Concomitantly, the number of branching points significantly increased, while end-point moving speed of microglial processes decreased (**Fig. 2e-h** [↗](#)), indicating state-dependent changes in microglial morphology as well as process motility under natural sleep-wake cycles. To provide a volumetric view of the dynamics of microglial morphology, we adapted our latest version of mTPM with an electrical tunable lens (ETL) capable of imaging variable focal planes (**Fig. 2i-o** [↗](#) and Supplementary Fig. 2) (Zong et al., 2021 [↗](#)). Multi-plane reconstruction revealed more extended process lengths and greater branch point numbers as compared to single-plane imaging as expected (**Fig. 2l, n** [↗](#)). The volume of surveillance for individual microglia changed from $1661 \pm 264.4 \mu\text{m}_3$ in the wake state to $3802 \pm 670.5 \mu\text{m}_3$ in REM state and $4616 \pm 324.6 \mu\text{m}_3$ in NREM state (**Fig. 2m** [↗](#)). Similar to results obtained with single-plane imaging, the end-point speed of processes decreased in REM and NREM compared to the wake state (**Fig. 2o** [↗](#)). While both REM and NREM had the same trends in change as compared to the wake state, significantly greater changes were associated with NREM than REM in terms of cell volume, branch length, branch number, and end-point speed (**Fig. 2l-o** [↗](#)). These results demonstrate differential microglial dynamic surveillance as the brain state switches among wakefulness, REM and NREM, and the state-dependent morphological and process-motility changes can be robustly reproduced with either single-plane or volumetric imaging. Intriguingly, microglia can even sense and respond to subtle differences between REM and NREM states. Because more extended microglia are thought to have greater surveillance ability (Kierdorf and Prinz, 2017 [↗](#)), our results support the notion that microglia during the NREM and REM periods tend to be more active in the opportune clearing of metabolic wastes accumulated during the wake states.

To this end, it is also instructive to compare and contrast REM/NREM state-dependent characteristics with those induced by anesthesia. In mice under isoflurane (1.2% in air) for 20-30 min, we found that microglial length, area, number of branching points (Supplementary Fig. 2) were similarly increased compared with the wake state, as was the case in NREM and REM. However, the end-point movement of processes exhibited opposite changes (Liu et al., 2019 [↗](#)), increasing with anesthesia but decreasing in sleep states (**Fig. 2h, o** [↗](#) and Supplementary Fig. 2). This result shows that microglia surveillance appears to differ qualitatively between sleep and anesthesia states.

Acute SD promotes contraction of microglial processes

Sleep loss is detrimental to brain function and results in structural plastic changes in nerve cells. Whether microglia dynamic surveillance alters after sleep loss remains controversial (Bellesi et al., 2017; Wang et al., 2020 [↗](#)). Therefore, we induced acute SD in mice by forcing the animals to exercise and interrupting sleep with a rotatory rod, starting at 9:00 am and lasting for six hours (zeitgeber time 2-8) (**Fig. 3a** [↗](#)). The duration of recovery sleep, especially NREM sleep, increased after SD (Supplementary Fig. 3). In the wake state, microglia typically exhibited an extended form with ramified, long, thin processes. Overt morphological changes occurred as early as 3h after SD, when microglia presented a contracted form bearing short and thick processes. By 6h, nearly all microglia cells were converted into a phagocytic form, assuming an amoeboid shape with few processes (**Fig. 3b-d** [↗](#)). We found that microglial length, area and number of branching points

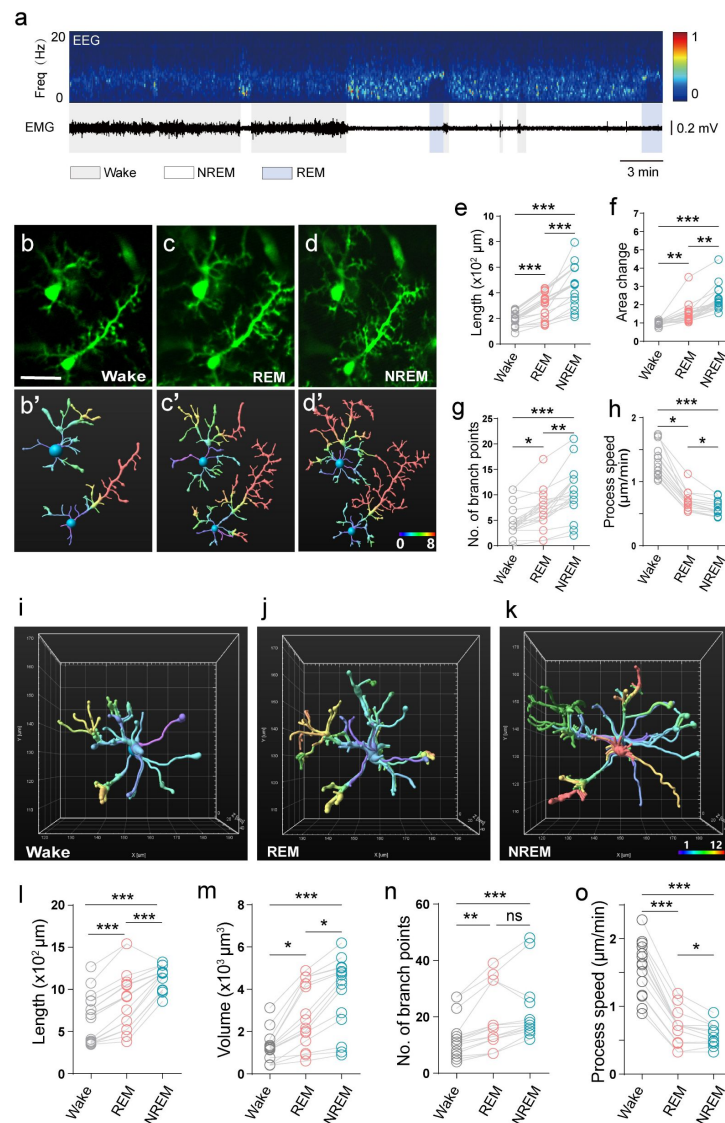


Fig. 2.

Microglial surveillance is state-dependent in the sleep-wake cycle.

(a) Representative EEG/EMG recordings showing the sleep-wake stage switch. Top, EEG power spectrogram (1-20 Hz). Middle, EMG trace. Bottom, brain states classified as wake (color code: gray), REM (blue), and NREM (white).

(b-d) Representative microglial morphological changes during the sleep-wake cycle.

(b'-d') Microglial morphology reconstructed from b-d using Imaris software.

(e-h) Microglial morphological parameters, length (e), area change (f), number of branch points (g), and process end point speed (h), all exhibited brain state-dependent dynamic change.

One-way ANOVA with Tukey's post-hoc test in **e**; Friedman test with Dunn's post-hoc test in **f-h**; $n = 20$ cells from 7 mice for each group (**e-g**), $n = 15$ cells from 6 mice for each group (**h**); $*P < 0.05$, $**P < 0.01$, $***P < 0.001$.

(i-o) 3D multi-plane imaging and reconstruction of microglial morphology. A 3D ETL lens was used to acquire multi-plane imaging ($220 \times 220 \times 40 \mu\text{m}_3$) at Z-intervals of $2 \mu\text{m}$, at a rate of 7.5 stacks/5 min.

(i-k) 3D reconstructed microglial morphology in wake (i), REM (j), and NREM (k) states, with corresponding time stamps shown at the bottom.

(l-o) Quantitative analysis of microglial length (l), volume change (m), number of branch points (n), and process motility (o) based on multi-plane microglial imaging. Data from 3D imaging corroborated state-dependent changes of microglial morphology in the sleep-wake cycle. Scale bars, $30 \mu\text{m}$.

One-way ANOVA with Tukey's post-hoc test in **l, o**; Friedman test with Dunn's post-hoc test in **m, n**; $n = 17$ cells from 7 mice for each group (**l-n**), $n = 15$ cells from 6 mice for each group (**o**); $*P < 0.05$, $**P < 0.01$, $***P < 0.001$.

decreased with the continuous SD (**Fig. 3e-h**). The same changes were observed in 3D reconstruction (Supplementary Fig. 3). Changes in the end-point speed of processes appeared to be biphasic, increasing at 3h SD and then declining below baseline level at 6h SD (**Fig. 3h**); the latter may result in part from the dramatic shrinkage of the processes. As the sleep pressure was eased during recovery sleep, morphological changes of microglia were partially reversed over a timescale of several hours (Supplementary Fig. 3). Thus, our real-time recording in live animals clearly demonstrates that SD stress induces microglia to assume an active state and conspicuous microglial response occurs early in SD.

NE changes in somatosensory cortex during the sleep-wake cycle

Next, we sought to determine a possible signaling mechanism(s) that incites microglia surveillance state-and stress-dependent changes. Among a number of candidates responsible for stress responses, norepinephrine (NE) has recently been shown to play a crucial role in microglia surveillance by responding to neuronal network activity (Liu et al., 2019) and partaking of synaptic plasticity (Stowell et al., 2019). Therefore, we combined mTPM imaging with EEG and EMG recording to determine whether NE levels in the somatosensory cortex fluctuate during the sleep-wake cycle to control microglia surveillance. We imaged extracellular NE dynamics in somatosensory cortical neurons with NE biosensor GRAB_{NE2m} (**Fig. 4a**), which was developed based on G protein receptors (Feng et al., 2019; Kjaerby et al., 2022). The biosensor was expressed on the plasma membrane of neurons to report the dynamic change of the extracellular NE. Dynamic changes of NE levels was evident in the somatosensory cortex during wake, NREM and REM states (**Fig. 4b**). NE levels reached higher levels during longer awakenings, while the lowest NE levels were detected during REM sleep (**Fig. 4b**) and the largest changes in NE levels were recorded occurred during the transition from REM to wake (**Fig. 4b**). During NREM sleep, the NE oscillated dynamically around a relatively low level (**Fig. 4b**), with brief increases that might be related to memory consolidation (Kjaerby et al., 2022). On average, mean levels of NE exhibited a brain-state dependent change, varying in a decreasing order from wake to NREM and to REM (**Fig. 4c**). These results are in general agreement with recent observations that NE fluctuates in a sleep-state dependent manner in the medial prefrontal cortex, and suggest that NE may play a crucial role in controlling the stage dependent microglial changes.

Role of the LC-NE signal in controlling the microglia dynamic surveillance

To determine possible involvement of NE in sleep state-and SD-induced microglia surveillance, we used the locus coeruleus (LC)-selective neurotoxin, N-(2-chloroethyl)-n-ethyl-2-bromobenzylamine (DSP4; applied 2 days before imaging) to ablate LC axons projecting into the cortex (Supplementary Fig. 4) (Del C. González et al., 1998) and characterized microglial surveillance with and without an LC-NE signal (**Fig. 5a**). DSP4 application led to a significant increase in sleep states, particularly NREM sleep (Supplementary Fig. 4), in agreement with previous studies (Del C. González et al., 1998). Meanwhile, the characteristics of NE dynamics in different brain states were also changed by DSP4 (Supplementary Fig. 4). Importantly, the sleep state-dependent changes in microglia surveillance were abolished altogether; neither microglial morphology nor process motility displayed any of the previously significant changes seen when the brain states cycled across wake-REM-NREM (**Fig. 5b-g**). Likewise, DSP4 treatment completely prevented SD-induced alteration of microglia surveillance (Supplementary Fig. 4). These results indicate an essential role for the LC-NE signal in sleep-and stress-dependent state modulation of microglia surveillance.

The effects of the neural modulator NE are mediated by two families of G-protein-coupled receptors, α and β -adrenergic receptors (ARs), each comprising several subtypes and it has been shown that β_2 AR stimulation rapidly induces microglia dynamic surveillance (Gyoneva and

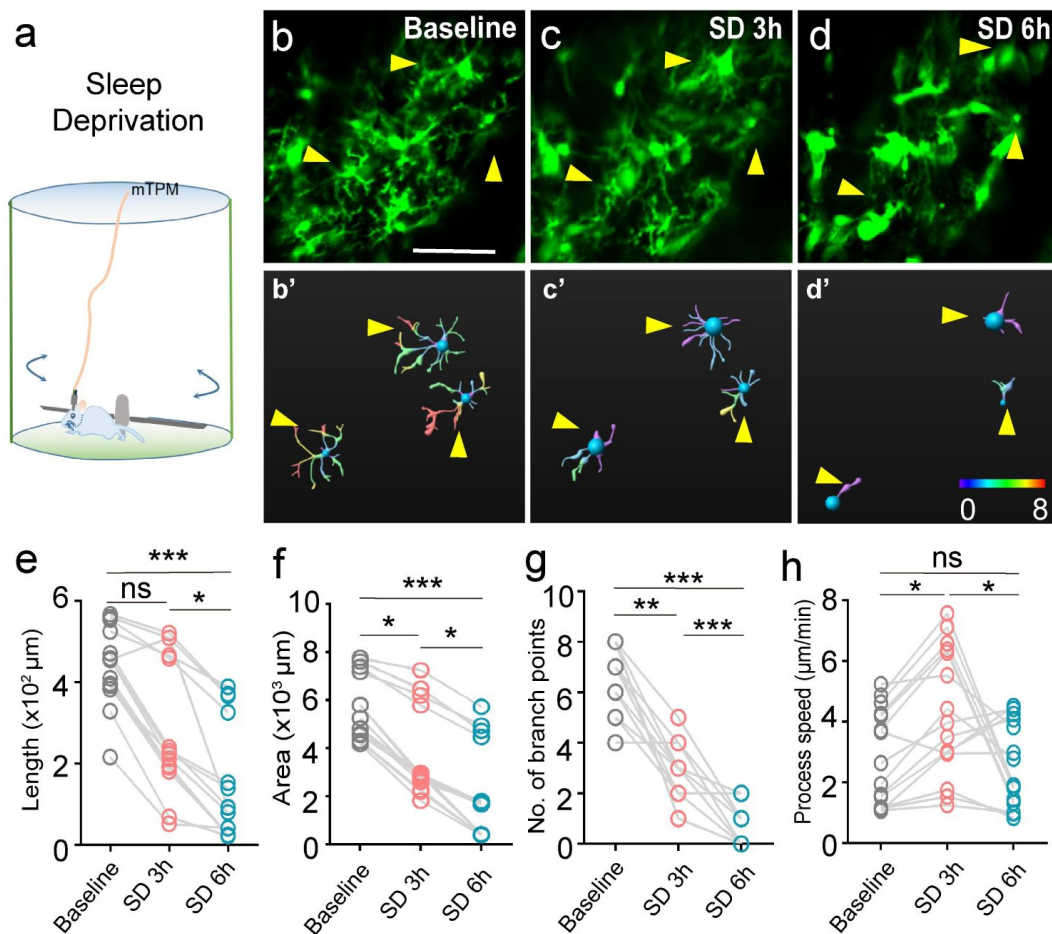


Fig. 3.

Changes of microglial surveillance in the state of sleep deprivation.

(a) Experimental setup for sleep deprivation. Sleep deprivation in mice was achieved by forcing them to exercise and interrupting their sleep with the rotation of a 46-cm rod (18 turns/min) in the chamber (diameter 50 cm).

(b-d) Microglial processes contracted after sleep deprivation (SD), baseline (b), SD 3h (c), and SD 6h (d). **(b'-d')** Morphological changes of microglia reconstructed using Imaris software. **b'-d'** correspond to **b-d**, respectively.

(e-h) Statistics for length (e), area (f), number of branch points (g), and process motility (h). Scale bar, 30 μm .

One-way ANOVA with Tukey's post-hoc test in **g**; Friedman test with Dunn's post-hoc test in **e, f, h**; $n = 15$ cells from 6 mice for each group; * $P < 0.05$, ** $P < 0.01$,

*** $P < 0.001$.

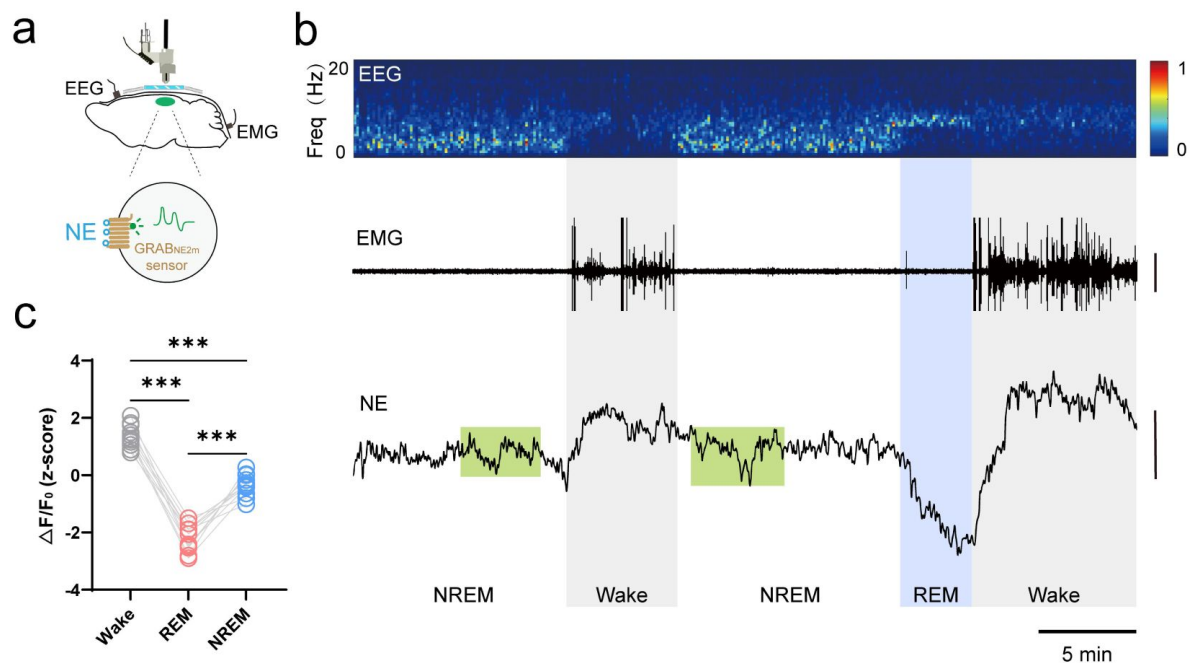


Fig. 4.

NE dynamics in mouse somatosensory cortex during the sleep-wake cycle.

(a) Schematic diagram depicting mTPM recording of extracellular NE indicated by the GRAB_{NE2m} sensor expressed in neurons.

(b) Representative traces of simultaneous recordings in the somatosensory cortex during the sleep-wake cycle in freely behaving mice. EEG and its power spectrogram (0-20 Hz); EMG (scale, 50 μ V); NE signals reflected by the z-score of the GRAB_{NE2m} fluorescence (scale, 2 z-score). The brain states are color-coded (wake, gray; NREM, white; REM, blue; NE oscillatory during NREM sleep, green).

(c) Mean extracellular NE levels in different brain states. Data from the same recording are connected by lines. *** $P < 0.001$, one-way ANOVA with Tukey's post-hoc test ($n = 11$ from 3 mice).

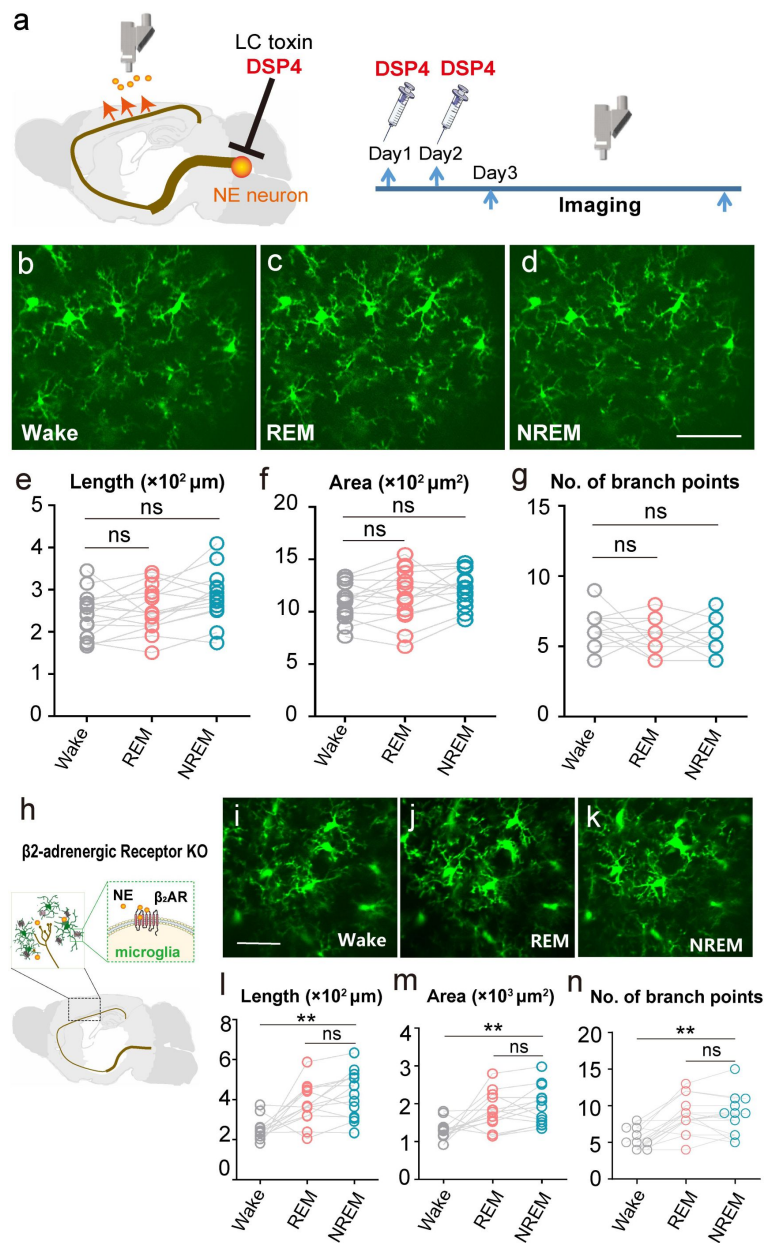


Fig. 5.

Microglial surveillance during natural sleep is controlled by LC-NE signal.

(a) Experimental setup: LC-selective neurotoxin DSP4 was used to destroy LC-NE neuronal axons.

(b-g) Lack of sleep/wake state-dependent microglial surveillance in LC-axon ablated animals. **b-d**: Representative mTPM images of microglia at different states. **e-g**: Statistics for microglial length (**e**), surveillance area (**f**), and number of branch points (**g**) in LC-axon ablated mice. One-way ANOVA with Tukey's post-hoc test in **e, f**; Friedman test with Dunn's post-hoc test in **g**; $n = 17$ cells from 6 mice for each group; ns, not significant.

(h) Schematic diagram for β_2 ARs on the plasma membrane of microglia in the cerebral cortex responding to NE released from axonal terminals projected from LC.

(i-k) State-dependent microglial surveillance during sleep-wake cycle in β_2 AR knockout mice. Representative microglial images (**i-k**) and statistics for microglial process length (**l**), surveillance area (**m**), and number of branch points (**n**) at different states in (CX3CR1-GFP $^{+/-}$; Adrb2 $^{-/-}$) mice. Scale bar, 30 μ m.

One-way ANOVA with Tukey's post-hoc test in **m**; Friedman test with Dunn's post-hoc test in **l, n**; $n = 15$ cells from 6 mice for each group; ns, not significant, $**P < 0.01$.

Traynelis, 2013 [↗](#); Liu et al., 2019 [↗](#); Stowell et al., 2019 [↗](#)). We, therefore, examined microglia surveillance in β_2 AR KO animals during natural sleep-wake cycles and under SD stress. Similar to LC axon ablation, β_2 AR KO disturbed wake-sleep homeostasis, as manifested by a lengthening of overall sleep duration (Supplementary Fig. 4). By contrast, removal of β_2 ARs failed to abolish dynamic changes in microglia when the brain state switched between sleep and wakefulness (**Fig. 5h-n** [↗](#)). Nonetheless, the ability of microglia to subtly distinguish between REM and NREM states was largely compromised (**Fig. 5h-n** [↗](#) and Supplementary Fig. 4), suggesting a partial contribution of β_2 AR signaling to LC-NE modulation of microglia surveillance during sleep-wake cycles. Regarding microglia responses to SD stress, we showed that microglial morphology and motility remained unchanged during the SD protocol in β_2 AR KO animals (Supplementary Fig. 4), revealing a predominant role for β_2 AR signaling in LC-NE modulation of microglia surveillance under SD stress.

Discussion

With the development of mTPM imaging and EEG/EMG recording in freely behaving mice, we now provide direct evidence that microglia surveillance in mouse somatosensory cortex is sleep-state dependent during natural sleep-wake cycles and that microglia contract from ramified into phagocytic forms a few hours after the onset of acute SD stress. Combined with pharmacological intervention and the use of NE biosensor and genetically manipulated animals, we showed that the LC-NE pathway mediates these state-dependent changes in microglia surveillance and that β_2 AR signaling is involved in various aspects of the microglial responses, extending recent reports that the same pathway underlies microglia surveillance in the context of neuronal network activity (Liu et al., 2019 [↗](#)) and synaptic plasticity (Stowell et al., 2019 [↗](#)).

That microglia surveillance is sleep state-dependent substantiates an intimate relationship between rapid, robust state-dependent microglia remodeling and the biology of the sleep-wake cycle (Feng et al., 2019; Frank and Heller, 2018 [↗](#); Xie et al., 2013 [↗](#)). It has been shown that microglia play a central role in the removal of metabolic waste and even cell debris from the brain (Huisman et al., 2022; Márquez-Ropero et al., 2020 [↗](#)). Our study shows that microglia in the sleeping brain have larger area and length, consistent with greater clearance ability in the sleeping state than the waking. Recently, it has been reported that, in head-fixed mice, microglial morphological complexity was decreased in NREM sleep than during wakefulness (Hristovska et al., 2022 [↗](#)). In consideration of the altered durations of both NREM and REM sleep (Hristovska et al., 2022 [↗](#)), the unnatural sleep state would lead to an increase in the microarousal state, and ultimately lead to a change in the structure of the sleep state, which may be the main reason for the difference in microglia behavior from our natural sleep. Furthermore, our results show that microglia surveillance significantly differs between REM and NREM sleep in the somatosensory cortex, likely reflecting metabolic and functional differences between the two states. More specifically, previous reports have proposed differential roles between the NREM and REM sleep for consolidation of different types of memories. For example, NREM sleep contributes to declarative memories, whereas REM sleep is important for procedural and emotional memories (Besedovsky et al., 2019 [↗](#); Rasch and Born, 2013 [↗](#)). Memory consolidation is closely related to synaptic plasticity, in which microglia play a very important role (Corsi et al., 2022; Tuan and Lee, 2019 [↗](#)). In this regard, our findings hint at the possibility that not all features of microglia-neuron interactions, including mode of action and localization, are equal in the consolidation of different types of memory.

It has been shown that a large proportion of Iba1-immunoreactive microglia with larger cell bodies and less ramified processes appear in hippocampal brain slices following 48-72h of sleep deprivation and in the frontal cortex after 4.5 days of chronic sleep restriction, whereas acute sleep deprivation for 8 hours had no effect on microglial morphology (Bellesi et al., 2017; Hall et al., 2020 [↗](#)). However, these findings have mostly been derived from slice staining. While the

present work support the general conclusion that microglia surveillance is responsive to SD stress, our real-time mTPM imaging *in vivo* also unmasked a progressive change in microglia surveillance that occurred as early as 3h after onset of SD and full conversion of microglia into the phagocytic form after 6h SD stress. This apparent disparity between the current and previous reports might reflect the complications arising from *ex vivo* sample preparation, because microglia surveillance is highly dynamic and sensitive to neural environmental changes, evidenced by its cyclic changes during the sleep-wake switch.

We have provided two lines of evidence that the LC-NE- β_2 AR axis is involved in microglial surveillance both during natural sleep-wake cycle and under SD stress. Ablation of LC-NE neuronal projection abolished and knockout of β_2 ARs markedly altered the manifestation of the state-dependence of microglia surveillance, while the latter recapitulated most, but not all, the phenotypes of the former. These findings are well supported by emerging evidence that NE is necessary for microglial activation (Bellesi et al., 2016 [DOI](#); Berridge et al., 2012 [DOI](#)). First, NE is known to be a key neurotransmitter that regulates sleep dynamics (Wang et al., 2020 [DOI](#)) and the dynamics of microglia (Liu et al., 2019 [DOI](#); Sugama et al., 2019 [DOI](#)). Second, it has been shown β_2 AR expressed at higher level in microglia than other CNS cell types (Zhang et al., 2014 [DOI](#)) and microglia dynamic surveillance rapidly responds to β_2 AR stimulation (Gyoneva and Traynelis, 2013 [DOI](#); Liu et al., 2019 [DOI](#); Stowell et al., 2019 [DOI](#)). Moreover, the morphology of microglia in somatosensory cortex is also regulated by neuronal activity, in a microglial β_2 AR-dependent manner (Liu et al., 2019 [DOI](#)). Our results also extend the classic view of NE regulation of sleep and wakefulness by revealing a prominent microglial component in the NE response. It should be noted that pan-tissue β_2 AR knockout animal model was used in the current study and it warrants future investigation to pinpoint specific roles of microglial β_2 AR in the brain-state dependent microglial responses.

In addition to NE, other potential modulators also present dynamic during sleep-wake cycle and may partake in the regulation of microglia dynamic surveillance. It has been reported that LC firing stops (Aston-Jones et al., 1981; Rasmussen et al., 1986 [DOI](#)), while inhibitory neurons, such as PV neurons and VIP neurons, become relatively active during REM sleep (Brécier et al., 2022 [DOI](#)). ATP level in basal forebrain is shown to be higher in REM sleep than NREM sleep (Peng et al., 2023 [DOI](#)). Similarly, adenosine level in the somatosensory cortex during REM sleep is higher than in NREM sleep (Response [figure 1](#) [DOI](#)). These considerations may explain the finding that β_2 AR knockout failed to abolish microglial responses to sleep state switch and SD stress altogether. A more complete understanding of the regulatory mechanisms of microglia surveillance will help to delineate more precisely roles of microglia in physiology and under stress.

In summary, by real-time *in vivo* mTPM imaging of microglia surveillance, we uncovered the state-dependence of microglia surveillance during natural sleep-wake cycles and a robust, early-onset response to SD stress. Both types of microglial dynamics are under the regulation of LC-NE with the involvement of β_2 ARs as one of its main effectors. These results highlight that microglia undergo rapid and robust remodeling of both morphology and function to fulfill multimodal roles complementary to those of neurons in the brain. In addition, the methodologies established in the present study may prove to be of broad application for future investigations of microglia dynamics and sleep biology in freely behaving animals.

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Author contributions

X.G. and H.C. conceived the project and supervised the research; X.G. designed and supervised the biological experiments. X.G., Z.Z. and X.C. performed the experiments and led data analysis and figure-making; X.Q. and W.G. contributed data analysis under the supervision of H.C., J.Z., and X.G.; X.G. and H.C. wrote the manuscript with inputs from L.Z. and H.F., while all authors participated in discussions and data interpretation.

Competing interests

The authors declare no competing interests.

Methods

Animals

All experimental protocols were carried out with the approval of the Institutional Animal Care and Use Committee of PKU-Nanjing Institute of Translational Medicine (Approval ID: IACUC-2021-023). Male mice, 2-3 months of age, were used in accordance with institutional guidelines. Cx3cr1-GFP (Jax, #021160) heterozygous mice were used to visualize microglia with miniature two-photon microscopy (mTPM). To ablate LC-NE neurons, DSP4 (Sigma, #C8417) solution was administered intraperitoneally (50 mg/kg) twice at 24-h intervals and a minimum of 48 h before mTPM imaging. Adrb2 KO (Jax, #031496) mice were crossed with Cx3cr1-GFP mice to generate (Cx3cr1-GFP^{+/+}; Adrb2^{-/-}) mice, which were used in experiments with β_2 -adrenergic receptor knock-out. Mice were housed with a standard 12h light/12h dark cycle and fed standard chow *ad libitum*.

Surgery and electrode implantation

All surgical procedures were done sterilely, and all animal-administered reagents were sterile. Mice were anesthetized with isoflurane (1.5% in air at a flow rate of 0.4 L/min) and maintained on a 37°C heating pad during surgery. The cerebral cortical region to be imaged was localized based on the stereotactic coordinates (somatosensory area, from bregma: anteroposterior, -1 mm, mediolateral, +2 mm) and marked with a fine black dot. The skull over the region of interest was thinned with a high-speed micro-drill under a dissection microscope. Drilling and application of normal saline were done intermittently to avoid overheating and hurting the underlying cerebral tissue. After removing the external layer of the compact bone and most of the spongy bone layer with the drill, the area continued to thinned until an expanded smooth area (~3 mm in diameter) was achieved, making sure the middle was thin enough to get high quality imaging. A drop of saline was applied and a 3-mm-diameter glass coverslip previously sterilized in 70% ethanol was placed on this window, then dental cement was applied around the glass coverslip. The mice were used in the experiment after 4 weeks of postoperative recovery.

500 nL of AAV9-hSyn-GRAB_{NE2m} (BrainVTA, #PT-2393) were injected into the somatosensory cortex (A/P, -1 mm; M/L, +2 mm; D/V, -0.3mm) at a rate of 50 nL/min. 3-mm-diameter glass coverslip were implanted and fixed on the somatosensory cortex. The mice were used in the experiment one month after postoperative recovery.

Epidural screw electrodes (diameter 0.8 mm) were implanted bilaterally on the opposite sides of the imaging window for constant EEG recording from the frontal (bregma: anteroposterior, +1.5 mm, mediolateral, +1.5 mm) and parietal cortex (anteroposterior, -2 mm; mediolateral, +2.5 mm). Electrodes were fixed to the skull with dental cement. Another pair of electrodes was inserted into the neck muscles for EMG recording. Then, a headpiece baseplate was attached to the skull with cyanoacrylate and reinforced with dental cement.

Simultaneous imaging and recording

After recovery from surgery, mice with a clear cranial window and typical EEG/EMG signals were selected for further experiments. With the mice head-fixed on the imaging stage, an mTPM imaging stack was first acquired using an mTPM (Transcend Vivoscope), and a proper field-of-view (FOV) with 5-10 microglial somata was located 50-100 μm beneath the pial surface. The holder of the mTPM was then sealed onto the baseplate over the coverslip on the head, and the mTPM could be repetitively mounted and dismounted to track the same population of microglia over different days (Supplementary Fig. 1).

Three models of mTPM were used for different experiments: FHIRM-TPM high-resolution model (Zong et al., 2017 [DOI](#)) with lens NA of 0.7 and FOV of $220 \times 220 \mu\text{m}$ for **Figs. 1 [DOI](#)-5 [DOI](#)**; headpiece weight was 2.13 g, and lateral and axial resolutions were 0.74 and 6.53 μm , respectively. FHIRM-TPM large FOV model (Zong et al., 2021 [DOI](#)) with a headpiece weight of 2.45 g, lens NA of 0.5, and FOV of $420 \times 420 \mu\text{m}$ for Supplementary Fig. 4, and lateral and axial resolutions were 1.13 and 12.2 μm , respectively; FHIRM-TPM 2.0 ETL model (Zong et al., 2021 [DOI](#)) in Supplementary Fig. 2, with headpiece weight of 4.3 g, Z-depth range of 45 μm , lens NA of 0.7, lateral resolution of 0.74 μm , axial resolution of 6.53 μm , and FOV of $220 \times 220 \mu\text{m}$. Images were acquired to visualize the microglial surveillance: time-lapse xy imaging stack at a frame rate of 5 FPS or multi-plane imaging stacks (20 planes at 2 μm intervals, 5 frames/s). Each model of mTPM was equipped with a water-immersion miniature objective and a 920 nm femtosecond fiber laser. Excitation for fluorescence imaging was achieved with 150-fs laser pulses (80 MHz) at 920 nm for GFP with a power of $\sim 25 \text{ mW}$ after the objective.

The EEG/EMG signals were recorded continuously and simultaneously by Vital Record (Kissei Comtec system), and mouse behavior was monitored with an infrared video camera. Mice were never disturbed when they were undergoing spontaneous sleep-wake cycles, feeding, or drinking.

Sleep deprivation

Sleep deprivation (SD) was achieved by forcing the mice to move continuously through power devices (Soft maze, #XR-XS108) (Pandi-Perumal et al., 2007 [DOI](#)). The experimental device consisted of a computer console, a horizontal rotating rod with a diameter of 46 cm and a round mouse cage of 50 cm. The rotating rod can rotate horizontally randomly (3 times/min) in clockwise and counterclockwise directions under the control of the computer. The mouse was placed in the cage one hour per day for one week for habituation. After baseline sleep recording, mice were sleep-deprived for the duration indicated in each experiment, starting at 9:00 AM and lasting up to 6 h into SD.

Anesthesia experiment

Time-lapsed image stacks acquired between 20 and 30 min after anesthesia (isoflurane, 1.5% in air at a flow rate of 0.4 L/min) were used to evaluate microglial dynamic surveillance under anesthetized conditions.

Quantification and statistical analysis

EEG/EMG data analysis

EEG and EMG signals were amplified and filtered as follows: EEG, high-pass filter at 0.1 Hz, low-pass filter at 35 Hz; and EMG, high-pass filter at 10 Hz, low-pass filter at 100 Hz. All signals were digitalized at 128 Hz and stored on a computer. As described previously (Bellesi et al., 2013 [DOI](#)), EEG power spectra were computed by a fast Fourier transform routine for 10 s epochs. Wake, nonrapid eye movement (NREM) sleep, and REM sleep were manually scored off-line (SleepSign, Kissei Comtec) in 10 s epochs according to standard criteria (Fogel and Smith, 2011 [DOI](#)). Epochs containing artifacts, predominantly during animal movement, were excluded from spectral analysis.

Microglial Morphological Analysis

In order to accurately quantify microglial morphometric changes in different brain states, wake and NREM states lasting longer than 1 min were chosen, among which the last 30 seconds of recording was selected, in conjunction with video interpretation, for further processing and analysis. Because they were of relatively short duration, REM states lasting longer than 30 seconds were used for analysis.

Image stacks were first processed offline using ImageJ software to correct for xy motion artifacts, and the time-lapse stack was aligned using the StackReg plugin. In some experiments, motion artifacts due to Z-level drift was further minimized by collecting multi-plane images over $\pm 20 \mu\text{m}$ depth at $2 \mu\text{m}$ intervals and then compressing them into a single xy projection to substitute for time-lapsed xy image. In this case, corresponding projections were generated over a 30-seconds window to substitute for time-lapsed xy images. A maximum-intensity projection (MIP) was then created from the stack for morphological analysis. The criteria to select microglia for analysis included a clearly identifiable soma and processes seen in the x-y plane.

Next, microglial morphological analysis was done using Imaris 9.5 software and the Filament module (Supplementary Fig. 1). The filament created in Imaris is a connected graph-based object consisting of vertices that are connected by edges.

Feature extraction by the Filament module yielded microglial features and parameters defined as the following:

Microglial process

Same a filament dendrite in module. Microglial processes form the main structure of the filament object. Processes extend from the microglial cell body and can undergo branching. A process graph is a sequence of vertices connected by edges.

Number of branch points

Same a filament branch depth in module. The process graph is a tree-based structure that has a root point. Branching depth is defined as the number of branches or bifurcations in the shortest path from the beginning point to a given point in the process graph.

Process length

Same a filament dendrite length in module, in reference to the sum of all process lengths within the entire microglial cell.

Microglial surveillance area

Same as filament dendrite area in module, in reference to the sum of all process areas within the entire microglial cell.

Microglial surveillance volume

Same as filament dendrite volume. In volumetric imaging, it is the sum of the volumes of all processes within the microglial cell.

Process end-point speed

Same as filament track speed. The speed is calculated using the “Track over time” function and given by the absolute track length change, whether extension or retraction, divided by the corresponding elapsed time.

NE signals analysis

To analyze NE signals data, we binned the raw data into 1 Hz and subtracted the background autofluorescence. $\Delta F/F_0$ were calculated by using the fluorescence signal itself. F_0 was defined as the mean value of the lowest fluorescence signal for each 60-second window in each recorded fluorescence signal. The z-score transformed $\Delta F/F_0$ was used for analysis. In order to accurately quantify the changes of NE signals in different brain states, wake and NREM states with a duration of more than 1 min were selected, in which the mean value of 30 seconds fluorescence signal at the end was selected as the NE signal in this state, which was further processed and analyzed combined with video interpretation. Since the duration of REM state is relatively short and it enters a low level at the end of REM, the mean value of fluorescence signal in the 30s before the transition from REM state to wake state is used to represent NE signal in REM state.

Statistical analyses

Significance levels indicated are as follows: $*P < 0.05$, $**P < 0.01$, $***P < 0.001$. All data are presented as mean $\pm$ SEM. All statistical tests used were two-tailed. Statistical significance was determined using both parametric (ANOVA) and non-parametric (Friedman) tests with post-hoc. All statistical testing was performed using GraphPad Prism 6.0 (GraphPad Software). No statistical methods were used to predetermine sample sizes, but our sample sizes were similar to those reported in previous publications (Davalos et al., 2005 [DOI](#); Nimmerjahn et al., 2005 [DOI](#)).

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Article and author information

Xiaochun Gu

Jiangsu Key Laboratory of Molecular and Functional Imaging, Department of Radiology, Zhongda Hospital, Medical School, Southeast University, Nanjing 210009, China, Key Laboratory of Developmental Genes and Human Diseases, Department of Histology Embryology, Medical School, Southeast University, Nanjing 210009, China, Research Unit of Mitochondria in Brain Diseases, Chinese Academy of Medical Sciences, PKU-Nanjing Institute of Translational Medicine, Nanjing 211899, China

For correspondence: xiaochun.gu@seu.edu.cn

Zhong Zhao

Institute of Basic Medical Sciences Chinese Academy of Medical Sciences, School of Basic Medicine Peking Union Medical College, Beijing 100005, China, Research Unit of Mitochondria in Brain Diseases, Chinese Academy of Medical Sciences, PKU-Nanjing Institute of Translational Medicine, Nanjing 211899, China

Xueli Chen

Research Unit of Mitochondria in Brain Diseases, Chinese Academy of Medical Sciences, PKU-Nanjing Institute of Translational Medicine, Nanjing 211899, China

Lifeng Zhang

Research Unit of Mitochondria in Brain Diseases, Chinese Academy of Medical Sciences, PKU-Nanjing Institute of Translational Medicine, Nanjing 211899, China

Huaqiang Fang

Research Unit of Mitochondria in Brain Diseases, Chinese Academy of Medical Sciences, PKU-Nanjing Institute of Translational Medicine, Nanjing 211899, China

Ting Zhao

Nanjing Raygenitm Biotech Co., Ltd, Nanjing 211899, China

Shenghong Ju

Jiangsu Key Laboratory of Molecular and Functional Imaging, Department of Radiology, Zhongda Hospital, Medical School, Southeast University, Nanjing 210009, China

Weizheng Gao

Academy of Advanced Interdisciplinary Study, College of Engineering, Peking University, Beijing 100871, China

Xiaoyu Qian

Academy of Advanced Interdisciplinary Study, College of Engineering, Peking University, Beijing 100871, China

Xianhua Wang

National Biomedical Imaging Center, State Key Laboratory of Membrane Biology, Peking-Tsinghua Center for Life Sciences, College of Future Technology, Peking University, Beijing 100871, China, Research Unit of Mitochondria in Brain Diseases, Chinese Academy of Medical Sciences, PKU-Nanjing Institute of Translational Medicine, Nanjing 211899, China

Jue Zhang

Academy of Advanced Interdisciplinary Study, College of Engineering, Peking University, Beijing 100871, China

Heping Cheng

National Biomedical Imaging Center, State Key Laboratory of Membrane Biology, Peking-Tsinghua Center for Life Sciences, College of Future Technology, Peking University, Beijing 100871, China, Research Unit of Mitochondria in Brain Diseases, Chinese Academy of Medical Sciences, PKU-Nanjing Institute of Translational Medicine, Nanjing 211899, China

For correspondence: chengp@pku.edu.cn

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Reviewer #1 (Public Review):

Microglia are increasingly recognized as playing an important role in shaping the synaptic circuit and regulating neural dynamics in response to changes in their surrounding environment and in brain states. While numerous studies have suggested that microglia

contribute to sleep regulation and are modulated by sleep, there has been little direct evidence that the morphological dynamics of microglia are modulated by the sleep/wake cycle. In this work, Gu et al. applied a recently developed miniature two-photon microscope in conjunction with EEG and EMG recording to monitor microglia surveillance in freely-moving mice over extended period of time. They found that microglia surveillance depends on the brain state in the sleep/wake cycle (wake, non-REM, or REM sleep). Furthermore, they subjected the mouse to acute sleep deprivation, and found that microglia gradually assume an active state in response. Finally, they showed that the state-dependent morphological changes depend on norepinephrine (NE), as chemically ablating noradrenergic inputs from locus coeruleus abolished such changes; this is in agreement with previous publications. The authors also showed that the effect of NE is partially mediated by β 2-adrenergic receptors, as shown with β 2-adrenergic receptor knock-out mice. Overall, this study is a technical tour de force, and its data add valuable direct evidence to the ongoing investigations of microglial morphological dynamics and its relationship with sleep. Nevertheless, microglial morphodynamics likely reflect the integrated influence of neighboring neuronal activities and neuromodulatory factors; the pan-tissue β 2AR knockout mouse model may also broadly affect the animal's physiology and sleep behavior. Therefore, future studies are needed to address the specific role of microglial β 2AR on its morphodynamics in sleep.

Reviewer #2 (Public Review):

The MS describes an approach to monitor microglial structural dynamics and correlate it to ongoing changes in brain state during sleep-wake cycles. The main novelty here is the use of miniaturized 2p microscopy, which allows tracking microglia surveillance over long periods of hours, while the mice are allowed to freely behave. Accordingly, this experimental setup would permit to explore long-lasting changes in microglia in more naturalistic environment, which were previously not possible to identify otherwise. The findings provide key advances to the research of microglia during natural sleep and wakefulness, as opposed to anesthesia. The main findings of the paper are that microglia increase their process motility and surveillance during REM and NREM sleep as compared to the awake state. The authors further show that sleep deprivation induces opposite changes in microglia dynamics- limiting their surveillance and size. The authors then demonstrate potential causal role for norepinephrine secretion from the locus coeruleus (LC) which is driven by beta 2 adrenergic receptors (b2AR) on microglia. '

The authors have nicely demonstrated and technically validated their main conclusions. In particular, they demonstrate the utility of miniaturized 2p imaging for long lasting imaging of microglia structural changes according to sleep state over the time course of hours. The authors have done a good job in addressing all my previous concerns and provide sound evidence for sleep state induced dynamics of microglia, which is modulated by NE and depends on b2AR.

One impressive point is the ability to longitudinally track the same microglial cells in the field of view for many hours, which is highly valuable and was impossible to achieve with head fixed imaging.

The authors support their observation by using a global b2AR KO mice, which ravel impaired microglial dynamics during sleep states.

While previous evidence supports high expression and function of b2AR in microglia, these receptors are expressed throughout the brain and periphery. Therefore, the authors correctly state that the current data they show, using global b2AR KO mice, cannot be used to state a direct effect on microglia dynamics and this would warrant future experiments with cell-specific genomic manipulation.

To summarize, the main conclusions of the paper are well validated and supported with the experimental layout and analysis.

Author Response

The following is the authors' response to the original reviews.

Response to the Referee Comments We would like to express our appreciation to the editor and the reviewers for their thoughtful comments and constructive suggestions on the manuscript. We agree with most of the comments and have carefully revised the manuscript accordingly. The revisions are highlighted in red font in the revised manuscript. Below are point-by-point responses to the referee's comments.

Public Reviews:

Reviewer #1 (Public Review):

Microglia are increasingly recognized as playing an important role in shaping the synaptic circuit and regulating neural dynamics in response to changes in their surrounding environment and in brain states. While numerous studies have suggested that microglia contribute to sleep regulation and are modulated by sleep, there has been little direct evidence that the morphological dynamics of microglia are modulated by the sleep/wake cycle. In this work, Gu et al. applied a recently developed miniature two-photon microscope in conjunction with EEG and EMG recording to monitor microglia surveillance in freely-moving mice over extended period of time. They found that microglia surveillance depends on the brain state in the sleep/wake cycle (wake, non-REM, or REM sleep). Furthermore, they subjected the mouse to acute sleep deprivation, and found that microglia gradually assume an active state in response. Finally, they showed that the state-dependent morphological changes depend on norepinephrine (NE), as chemically ablating noradrenergic inputs from locus coeruleus abolished such changes; this is in agreement with previous publications. The authors also showed that the effect of NE is partially mediated by β 2-adrenergic receptors, as shown with β 2-adrenergic receptor knock-out mice. Overall, this study is a technical tour de force, and its data add valuable direct evidence to the ongoing investigations of microglial morphological dynamics and its relationship with sleep. However, there are a number of details that need to be clarified, and some conclusions need to be corroborated by more control experiments or more rigorous statistical analysis. Specifically:

1. *The number of branch points per microglia shown here (e.g., Fig. 2g) is much lower than the values of branch points in the literature, e.g., Liu T et al., Neurobiol. Stress 15: 100342, 2021 (mouse dmPFC, IHC); Liu YU et al., Nat. Neurosci. 22: 1771-81, 2019 (mouse S1, in vivo 2P imaging). The authors need to discuss the possible source of such discrepancy.*

Thank you for raising this important point. Two reasons may account for this difference. Firstly, the difference in the definition of branch points in the software. Liu YU et al. used the Sholl analysis of image J software to analyze the number of branch points of microglia. Sholl analysis defines the number of branch points as the number of crossings between branches and concentric circles of increasing radii. We reconstructed microglia morphology using Imaris, a software that defines branching points based on the number of bifurcation points. The number of bifurcations calculated represents the number of microglia branch points. Secondly, this and previous studies found that more branching points present in the state of anesthesia. The morphological characteristics of microglia in head-fixed mice under anesthesia was reported by Liu T et al. and the microglia reconstruction results presented by the authors are indeed more complex than ours. In short, this is an aspect that we have been

paying attention to, and the main reasons for this difference may lie in the definition of branch points, analysis methods and related choice of thresholds. True differences in brain states and the heterogeneity of microglia in different brain regions may also contribute to the apparent discrepancy.

1. *Microglia process end-point speed (Fig. 2h, o): here the authors show that the speed is highest in the wake state and lowest in NREM, which agrees with the measurement on microglia motility during wakefulness vs NREM in a recent publication (Hristovska I et al., Nat. Commun. 13: 6273, 2022). However, Hristovska et al. also reported lower microglia complexity in NREM vs wake state, which seems to be the opposite of the finding in this paper. The authors need to discuss the possible source of such differences.*

This is also an important point. Hristovska et al. reported the morphodynamic characteristics of microglia during wakefulness and NREM sleep. It is worth noting that the sleep state of the mice in their experiments was unnatural due to the head fixation and body limitations, the duration of NREM sleep (sleep stability) being quite different from the NREM sleep analyzed under natural sleep. The limitations of this approach are also discussed by Hristovska et al. “Even though sleep episodes were, as anticipated, shorter than those observed in freely moving animals, changes in neuronal activity characteristic of NREM sleep were monitored by EEG recordings, and changes in morphodynamics were observed during single episodes. Several episodes of REM sleep were detected, but they were too short and rare to be analyzed reliably.” The unnatural sleep state would lead to an increase in the microarousal state, and ultimately a change in the structure of the sleep state, which may be the main reason for the difference in microglia behavior from our natural sleep. We have discussed this in the revised manuscript. Please see line 292298.

1. *Fig. 3: the authors used single-plane images to analyze the morphological changes over 3 or 6 hours of SD, which raises the concern that the processes imaged at the baseline may drift out of focus, leading to the dramatic reduction in process lengths, surveillance area, and number of branch points. In fact, a previous study (Bellesi M et al., J. Neurosci. 37(21): 5263-73, 2017) shows that after 8 h SD, the number of microglia process endpoints per cell and the summed process length per cell do not change significantly (although there is a trend to decline). The authors may confirm their findings by either 3D imaging in vivo, or 3D imaging in fixed tissue.*

Three lines of evidence indicate that microglia morphology changes in Fig 3 are due to SD, rather than variations in the focal plane. First, our single-plane images were quite stable over 3 or 6 hours of SD, though occasional reversible drifts might happen due to sudden motions. Second, per your suggestion, further experiments and analysis of 3D imaging were performed to monitor microglia dynamics during sleep deprivation. The new result is shown in revised Fig. S3 C-D: the length of microglia branches and the number of branching points were significantly reduced after SD, in agreement with the results of single-plane imaging. Furthermore, we detected no significant difference in microglia branching characteristics during 6h sleep deprivation in $\Delta 2AR$ KO mice (Fig.S4), and this indirectly affirmed that singleplane imaging is stable enough for detecting true changes in branching during SD.

1. *Fig. 4b: the EEG and EMG signals look significantly different from the example given in Fig. 2a. In particular, the EMG signal appears completely flat except for the first segment of wake state; the EEG power spectrum for REM appears dark; and the wake state corresponds to stronger low frequency components (below ~ 4 Hz) compared to NREM, which is the opposite of Fig. 2a. This raises the concern whether the classification of sleep stage is correct here.*

Thank you for insightful comments. We carefully examined the behavioral video of Figure 4b, there were occasionally microarousal events indicated by slow head rotation during NREM sleep, while the companion EMG signals were completely flat, which is atypical during sleep wake cycle. The microarousal events were not excluded from sleep, which makes this set of data unrepresentative and contrary to Fig.4b. In our revised manuscript, we replaced it with more representative data that can clearly and consistently distinguish between different brain states in mice on EMG and EEG. Please see revised Fig.2a, page 34; revised Fig.4b, page 37.

1. *Fig. 4 NE dynamics. • How long is a single continuous imaging session for NE? • When monitoring microglia surveillance, the authors were able to identify wake or NREM states longer than 15 min, and REM states longer than 5 min. Here the authors selected wake/NREM states longer than 1 min and REM states longer than 30 s. What makes such a big difference in the time duration selected for analysis? • Also, the definition of F0 is a bit unclear. Is the same F0 used throughout the entire imaging session, or is it defined with a moving window?*

A single continuous session of NE imaging usually took about 1 hour. Subsequent analysis was performed on imaging data from each recording that included wake, NREM sleep, and REM sleep. Because of the different time scales of microglia morphological dynamic (relatively slow) and NE signals (fast), we used different time windows in the previous analysis in the previous version of the manuscript.

Per your suggestion, we have now set the same time window selection criteria for both microglia morphological and NE dynamic analysis: for wake and NREM sleep durations longer than 1 minute, and REM sleep durations longer than 30 seconds. We updated the Methods and all statistics in related figures, please see line 151-154, 481-485, 490-492; Fig. 2e-g and 2l-n, page 34. F0 definition is now explained in the Methods section. Please see line 521-522.

1. *Fig. 5b: how does the microglia morphology in LC axon ablation mice compare with wild type mice under the wake state? The text mentioned "more contracted" morphology but didn't give any quantification. Also, the morphology of microglia in the wake state (Fig. 5b) appears very different from that shown in Fig. S3C1 (baseline). What is the reason?*

The morphology of microglia is indeed heterogeneous and variable, affected by factors including brain state, brain region, microenvironmental changes, along with animal-to-animal difference. We didn't perform the microglia morphology comparison between the LC axon ablation mice and wild type mice and, in view of this, we removed the description of "more contracted morphology" from the main text. It should also be noted that, as we primarily focused on changes of a microglia in different states over time by selfcomparison, we minimized possible effects of heterogeneity in microglia morphology on our conclusions.

1. *The relationship between NE level and microglia dynamics. Fig. 4C shows that the extracellular NE level is the highest in the wake state and the lowest in REM. Previous studies (Liu YU et al., Nat. Neurosci. 22(11):1771-1781, 2019; Stowell RD et al., Nat. Neurosci. 22(11): 1782-1792, 2019) suggest that high NE tone corresponds to reduced microglia complexity and surveillance. Hence, it would be expected that microglia process length, branch point number, and area/volume are higher in REM than in NREM. However, Fig. 2l-n show the opposite. How should we understand this ?*

Your point is well-taken. On the one hand, our data clearly showed that NE is critically involved in the brain state-dependent microglia dynamic surveillance, with evidence from

the ablation of the LC-NE projection and from the $\beta 2AR$ knockout animal model.

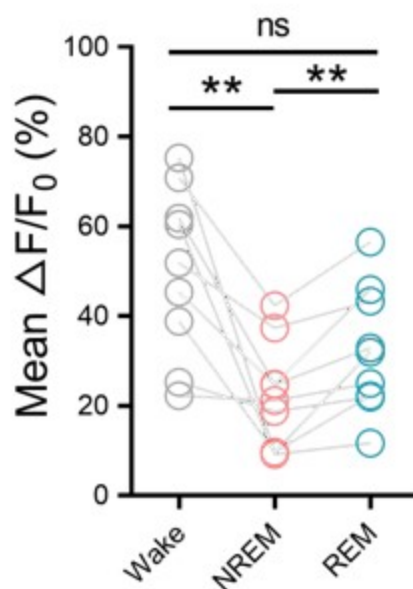
On the other hand, we also understand that NE is not the sole determinant, so the relationship between the NE level and the complexity and surveillance may not be unique.

In this regard, other potential modulators also present dynamic during sleepwake cycle and may partake in the regulation of microglia dynamic surveillance. previous studies (Liu YU et al., 2019; Stowell RD et al., 2019) have shown that microglia can be jointly affected by surrounding neuronal activity and NE level during wake. It has been reported that LC firing stops (Aston-Jones et al., 1981; Rasmussen et al., 1986), while inhibitory neurons, such as PV neurons and VIP neurons, become relatively active during REM sleep (Brécier et al., 2022). ATP level in basal forebrain is shown to be higher in REM than NREM (Peng et al., 2023). In addition, our own preliminary result (Author response image 1) also showed a higher adenosine level in REM than NREM in somatosensory cortex. Last but not the least, we found that $\beta 2AR$ knockout failed to abolish microglial responses to sleep state switch and SD stress altogether.

In brief, microglia are highly sensitive to varied changes in the surrounding environment, and many a modulator may participate in the microglia dynamic during sleep state. This may underlie the microglia complexity difference between REM and NREM. Future investigations are warranted to delineate the signal-integrative role of microglia in physiology and under stress. We have discussed the pertinent points in the revised manuscript. Please see line 343-354.

Author response image 1.

Extracellular adenosine levels in somatosensory cortex in different brain states. AAV2/9-hSyn-GRABAdo1.0 (Peng W. et al., Science. 2020) was injected into the somatosensory cortex (A/P, -1 mm; M/L, +2 mm; D/V, -0.3 mm). Data from the same recording are connected by lines. n = 9 from 3 mice.



Reviewer #2 (Public Review):

The manuscript describes an approach to monitor microglial structural dynamics and correlate it to ongoing changes in brain state during sleep-wake cycles. The main novelty

here is the use of miniaturized 2p microscopy, which allows tracking microglia surveillance over long periods of hours, while the mice are allowed to freely behave. Accordingly, this experimental setup would permit to explore long-lasting changes in microglia in a more naturalistic environment, which were previously not possible to identify otherwise. The findings could provide key advances to the research of microglia during natural sleep and wakefulness, as opposed to anesthesia. The main findings of the paper are that microglia increase their process motility and surveillance during REM and NREM sleep as compared to the awake state. The authors further show that sleep deprivation induces opposite changes in microglia dynamics- limiting their surveillance and size. The authors then demonstrate potential causal role for norepinephrine secretion from the locus coeruleus (LC) which is driven by beta 2 adrenergic receptors (b2AR) on microglia. However, there are several methodological and experimental concerns which should be addressed.

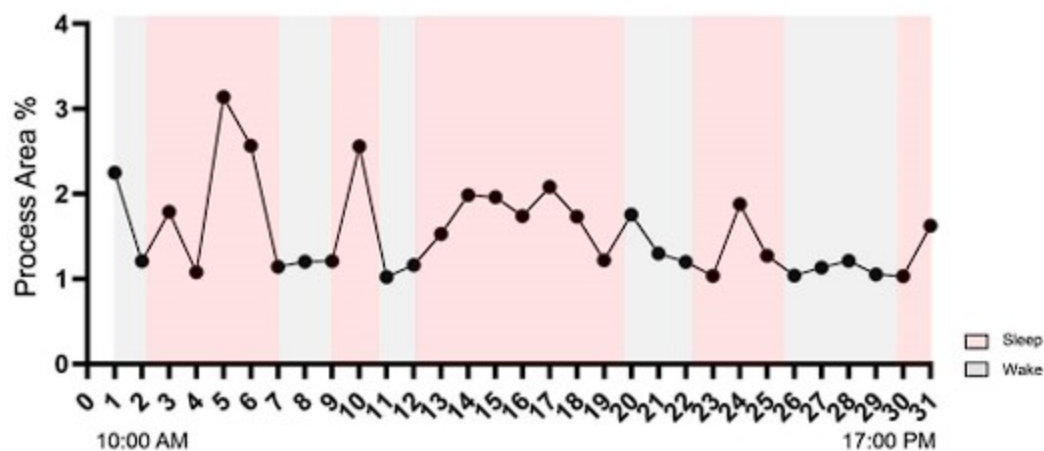
The major comments are summarized below:

1. The main technological advantage of the 2p miniaturized microscope is the ability to track single cells over sleep cycles. A main question that is unclear from the analysis and the way the data is presented is: are the structural changes in microglia reversible? Meaning, could the authors provide evidence that the same cell can dynamically change in sleep state and then return to similar size in wakefulness? The same question arises again with the data which is presented for anesthesia, is this change reversible?

As revealed by long-term free behavioral mTPM imaging, the brain-state-dependent morphological changes in microglia were reproducible and reversible. Author response image 2 shows that microglia displayed reversible dynamic changes during multiple rounds of sleep-wake transition. Author response image 3 shows that microglia dynamics induced by anesthesia also exhibited reversibility.

Author response image 2.

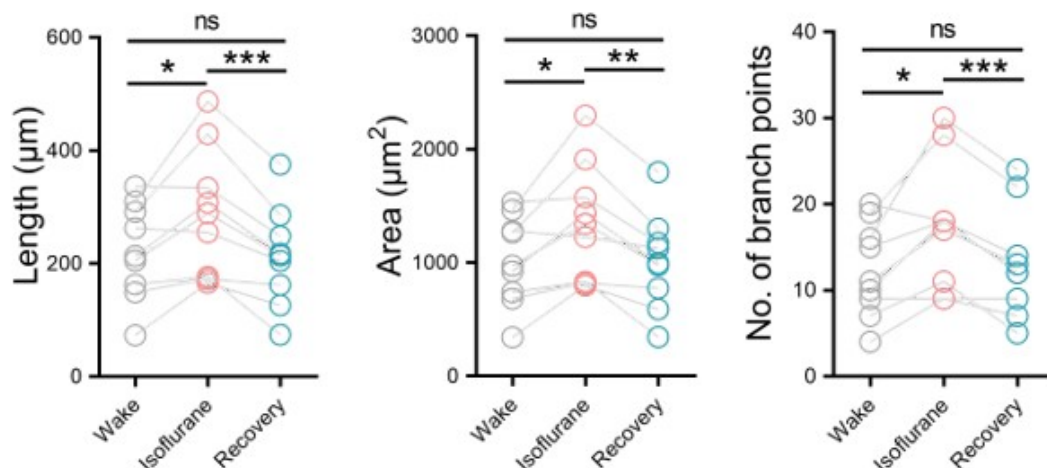
Long-term tracking of microglia process area in different brain states. Data analysis used 8 cells. Data total of 31 time points were selected from in vivo imaging data and were used to characterize the morphological changes of microglia over a continuous 7-hour period.



Author response image 3.

Reversible changes of microglial process length, area, number of branch points under anesthesia. Wake group: 30 minute-accommodation to new environment; Isoflurane group:

1.5% in air applied at a flow rate of 0.4 L/min for 30 minutes; Recovery group: 30 minutes after recovery from anesthesia. n = 9 cells from 3 mice for each group.



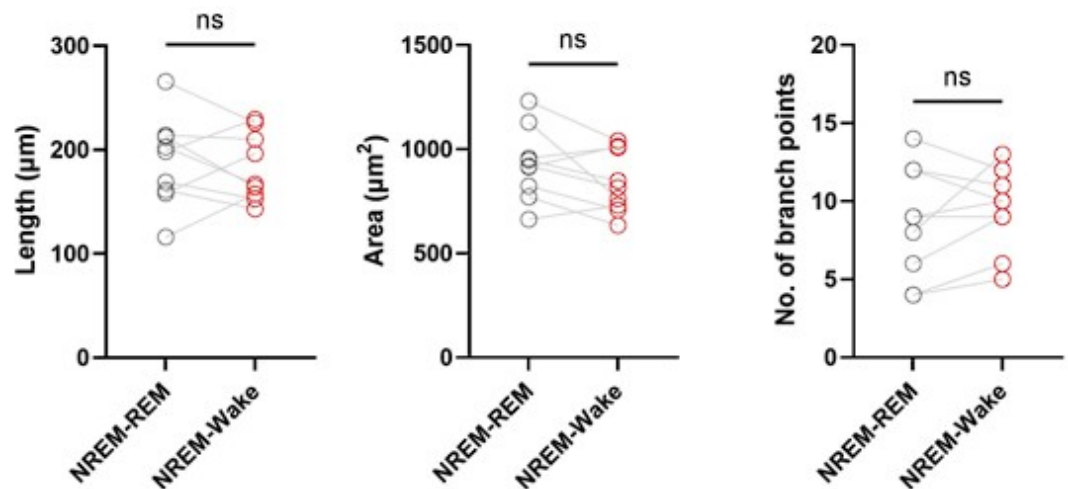
1. The binary comparison between brain states is misleading, shouldn't the changes in structural dynamics compared to the baseline of the state onset? The authors method describes analysis of the last 5 minutes in each sleep/wake state. However, these transitions are directional- for instance, REM usually follows NREM, so the description of a decrease in length during REM sleep could be inaccurate.

As you know, the time scale of microglia morphological dynamic is relatively slow, so we analyzed the microglia morphological dynamic of the last part (30s in the revised manuscript) of each state instead of the state onset, allowing time for stabilization of the microglia response to inter-state transition.

Further, we compared microglia dynamic between two NREM groups transiting to different subsequent states: group1 (NREM to REM) vs group2 (NREM to Wake). This precaution was to exclude the directional effect of state transitions. Our results showed that there was no difference in microglial length, area, number of branching points between the two NREM groups (Author response image 4), indicating that the last 30s of each NREM was not affected by its following state and that it's reasonable to perform binary comparison.

Author response image 4.

Microglial morphological length, area change, and number of branch points of the last 30s of NREM sleep followed by REM or Wake. n = 9 cells from 3 mice for each group.



1. Sleep deprivation- again, it is unclear whether these structural changes are reversible. This point is straightforward to address using this methodology by measuring sleep following SD. In addition, the authors chose a method to induce sleep deprivation that is rather harsh. It is unclear if the effect shown is the result of stress or perhaps an excess of motor activity.

We adopted the method of forced exercise as it has been commonly used for sleep deprivation (Pandi-Perumal et al., 2007; Nollet M et al., 2020), though it does have the potential limitation of excess of motor activity.

In light of your comments and suggestion, we presented new data demonstrating that sleep duration of the mice, mostly NREM sleep, increased compensatively (ZT9-10) after the 6-hour sleep deprivation (ZT2-8) (revised Fig. S3B). This result shows that sleep deprivation indeed increase sleep pressure in the mice. As the sleep pressure was eased during recovery sleep, morphological changes of microglia were reversed over a timescale of several hours (revised Fig. S3 E-J).

1. The authors perform measurements of norepinephrine with a recently developed GRAB sensor. These experiments are performed to causally link microglia surveillance during sleep to norepinephrine secretion. They perform 2p imaging and collect data points which are single neurons, and it is unclear why the normalization and analysis is performed for bulk fluorescence similar to data obtained with photometry.

We did not perform single-neuron analysis for two reasons. First, our experimental conditions, e.g., the expression of the NE indicator and the control of imaging laser intensity, did not yield sufficient signal-to-noise to clearly discriminate individual neurons with two-photon imaging. Second, NE signal may play a modulatory role, and fluorescence changes appeared to be global, rather than local or cell-specific. Therefore, we analyzed fluorescence changes in different brain states over the whole field-of-view in Fig. 4, rather than at the subregional or single-cell level.

1. *The experiments involving b2AR KO mice are difficult to interpret and do not provide substantial mechanistic insight. Since b2AR are expressed throughout numerous cell types in the brain and in the periphery, it is entirely not clear whether the effects on microglia dynamics are direct. The conclusion and the statement regarding the expression of b2AR in microglia is not supported by the references the authors present, which simply demonstrate the existence and function of b2AR in microglia. In addition, these mice show significant changes in sleep pattern and increased REM sleep. This could account for reasons for the changes in microglia structure rather than the interpretation that these are direct effects.*

To summarize, the main conclusions of the paper require further support with analysis of existing data and experimental validation.

Previous studies have revealed that norepinephrine (NE) has a modulating effect on microglial dynamics through β 2AR pathway (Stowell RD et al., 2019; Liu YU et al., 2019). Stowell et al. and Liu et al. use in vivo two-photon imaging to demonstrate that microglia dynamics differ between awake and anesthetized mice and to highlight the roles of NE and β 2AR in these states (Gyoneva S et al., 2013; Stowell RD et al., 2019; Liu YU et al., 2019). To evaluate the direct effect of β 2AR on microglial dynamics, Stowell et al. administered the β 2AR agonist clenbuterol to anesthetized mice and found that this decreased the motility, arbor complexity, and process coverage of microglia in the parenchyma (Stowell RD et al., 2019). Inhibition of β 2AR by antagonist ICI-118,551 in awake mice recapitulated the effects of anesthesia by enhancing microglial arborization and surveillance (Stowell RD et al., 2019). In addition, it has been shown microglia expressed higher numbers of β 2ARs than any other cells in the brain (Zhang et al., 2014).

To this end, our current work provided new evidence to support the involvement of the LC-NE- β 2AR axis in modulating microglia dynamics both during natural sleep-wake cycle and under SD stress. While we were aware the limitation of using pan-tissue β 2AR knockout model that precluded us from pinpointing role of microglial β 2AR, it is safe to state that β 2-adrenergic receptor signaling plays a significant role in the sleep-state dependent microglia dynamic surveillance, based on the present and previous data.

We have discussed this in the revised manuscript. Please see line 324-354. As you suggested, we added references to support the statement regarding the expression of β 2AR in microglia (please see line 333).

Recommendations for the authors: please note that you control which, if any, revisions, to undertake

Reviewer #1 (Recommendations For The Authors):

Some technical details need to be clarified. Also, please double-check for typos.

1. *In vivo imaging preparation: how long is the recovery time between window/EEG implantation surgery and imaging/recording?*

Imaging data were collected one month after the surgery. We have added descriptions to the methods section of the revised manuscript. Please see line 419.

1. *Statistical analysis: the authors used t-test or ANOVA without first checking whether the data pass the normality test. If the data does not follow a normal distribution, nonparametric tests would be more appropriate.*

Per your suggestion, we performed the test of statistical significance using parametric (ANOVA) if past the normality test, or the non-parametric (Friedman) tests for non-normal data. Please see line 533-535.

1. *Fig. 1b needs a minor change. In the figure, the EMG electrodes appear to be connected to the brain as well.*

We have corrected this oversight. Thank you.

1. *Fig. 1c: it would be helpful to give examples of raw EEG and EMG traces for REM and NREM separately.*

Raw traces are now shown as suggested. Please see Fig. 1c, page 32.

1. *Fig. 1h: is each data point one microglia or one end-point?*

In Fig. 1h, each data represents the average speed of all branches of one microglia, not one end-point.

1. Sleep deprivation starts at 9 am. What time corresponds to Zeitgeber Time 0 (ZT0, the beginning of the light phase)?

We now clarified that 9 am corresponds to Zeitgeber time 2. Please see line 196.

1. Line 61: the authors referred to Ramon y Cajal's original suggestion that microglia dynamics are coupled to the sleep-wake cycle. However, the cited paper only indicates that Cajal suggested a role of astrocytes in the sleep-wake cycle, not microglia. In addition, there is a typo in the line: there should be a space between "Ramon" and "y" in Cajal's name.

We have updated the statement and reference literature to point out the microglia's involvement in the sleep-wake cycle. The typo was corrected. Please see line 64-65.

1. *Fig. S3B: As each group has only 3 mice, it is unclear how t-test can yield $p < 0.01$ or even 0.001.*

We checked the original data again and it was correct. This small p-values may be due to the small intra-group difference of control group.

1. *Line 251-253, "Figure 4h-n" should be "Figure 5h-n"?*

We have revised it. Please see line 265-266.

1. *Fig. 5h: the receptor should be "adrenergic receptor", not "adrenal receptor".*

We changed the term to "adrenergic receptor". Please see Fig 5h.

1. Fig. 5g, n: the number of data points is apparently less than the sample size given in the figure legend. Perhaps some data points have exactly the same value so they overlap? The authors may consider plotting identical values with a slight shift so that the number of data points shown matches the actual sample size, to avoid confusion.

Yes, we have added small jitters so different data points can be seen to avoid confusion. Please see Fig. 5n.

1. There are some typos (e.g., Line 217, "he" should be "the") and some incomplete references (e.g., [13], [22], [34], [35] lack volume and page number, [15] and [39] lack publisher information). Some references have inconsistent formats (e.g., "Journal of Neuroscience" is sometimes abbreviated and sometimes not). Please correct these.

We have corrected these oversights. Please see references, page 27.

Reviewer #2 (Recommendations For The Authors):

Major issues:

1. Re-analyze the data in a manner that allows to follow and compare the same cells over different state transitions. This is necessary to evaluate the reversibility of microglia structure. In addition, consider analysis of the change from the beginning to the end of each state.

As shown in response figure 2, microglia dynamics were reversible during multiple rounds of sleep-wake transition.

1. It would be nice to see the raw data obtained over time, at least for Figure 1, before offline correction of movement to evaluate the imaging quality and level of drift during imaging.

We agree to your good suggestion. Please see the supporting material video.

1. It would be helpful to add an analysis of the percent time spent in each state for the 10 hour recordings.

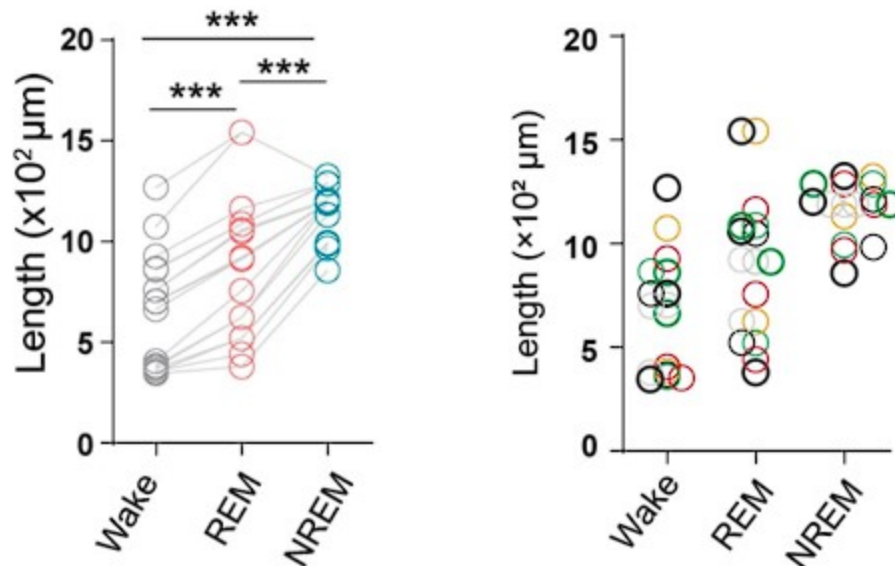
Advice has been adopted. Please see revised Fig. S4C.

1. In Figure 2 the results are from 15 cells from several animals. How much do the results vary between mice? It will be helpful to show if this varies between different mice by labeling cells from each mouse differently.

In Author response image 5, in which we have labeled the distribution of data points from seven mice, there was mixed distribution of data from different animals at each brain state, but no clear animal-to-animal difference.

Author response image 5.

Quantitative analysis of microglial length based on multi-plane microglial imaging. n = 17 cells from 7 mice for each group. In right panel, each color codes data from the same animal.



1. SD- please add some quantification for sleep and EEG to show that the manipulation really caused sleep deprivation. To address the confound of forced movement and stress, it might be helpful to add quantification of movement compared to an undisturbed wakefulness.

We have added related data (revised Fig. S3B), as suggested. Please see line 196-197.

1. The DSP4 application should be also performed with NE measurements to verify the specific of the NE signal measured as well as the DSP4 toxin.

Following your suggestion, we have added DSP4 data in revised Fig. S4B.

1. Some suggested refined experiments for the b2AR KO are: a- A conditional b2AR KO in microglia, as cited in the work. b- Local application of a b2 blocker during SD. c- Imaging of NE dynamics in the b2 animals. If NE dynamics during natural sleep cycle are perturbed, then this suggests upstream mechanisms rather than direct microglia effects as suggested by the authors.

We agree that the current study cannot pinpoint a direct effect of microglia harbored β 2AR. We have discussed this limitation in the revised manuscript.

Please see line 324-354.

Minor:

1. Typo on page 4 (microcopy instead of microscopy).

It was corrected. Please see line 87.

1. Typo page 11- 'and he largest changes in NE' - supposed to be 'the'.

We have corrected these mistakes. Please see line 228.

1. Fig. 4- there are several units missing in the figure in panel b: the top is Hz, but what does the color bar indicate exactly? 2 what? both for theta/delta and for NE. We have modified this figure and legend for clarity. Please see Fig. 4, page 37.

1. Bottom of page 12- referring to figure 4 but talking about figure 5.

The typo was corrected. Please see line 265-266.

Reference

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